### Example 3.31

Discuss the nature of the roots of the following polynomials:

(i) $x^{2018}+1947x^{1950}+15x^8+26x^6+2019$ (ii) $x^5-19x^4+2x^3+5x^2+11$

### Solution

Let $P(x)$ be the polynomial under consideration.

(i) The number of sign changes for $P(x)$ and $P(-x)$ are zero and hence it has no positive roots and no negative roots. Clearly zero is not a root. Thus the polynomial has no real roots and hence all roots of the polynomial are imaginary roots.

(ii) The number of sign changes for $P(x)$ and $P(-x)$ are 2 and 1 respectively. Hence it has at most two positive roots and at most one negative root.Since the difference between number of sign changes in coefficients of $P(-x)$ and the number of negative roots is even, we cannot have zero negative roots. So the number of negative roots is 1. Since the difference between number of sign changes in coefficient of $P(x)$ and the number of positive roots must be even, we must have either zero or two positive roots. But as the sum of the coefficients is zero, 1 is a root. Thus we must have two and only two positive roots. Obviously the other two roots are imaginary numbers. ■

## EXERCISE 3.6

1. Discuss the maximum possible number of positive and negative roots of the polynomial equation $9x^9-4x^8+4x^7-3x^6+2x^5+x^3+7x^2+7x+2=0$.
2. Discuss the maximum possible number of positive and negative zeros of the polynomials $x^2-5x+6$ and $x^2-5x+16$. Also draw rough sketch of the graphs.
3. Show that the equation $x^9-5x^5+4x^4+2x^2+1=0$ has atleast 6 imaginary solutions.
4. Determine the number of positive and negative roots of the equation $x^9-5x^8-14x^7=0$.
5. Find the exact number of real zeros and imaginary of the polynomial $x^9+9x^7+7x^5+5x^3+3x$.

## EXERCISE 3.7

**Choose the correct or the most suitable answer from the given four alternatives :**

1. A zero of $x^3+64$ is
   (1) 0 (2) 4 (3) $4i$ (4) -4
2. If $f$ and $g$ are polynomials of degrees $m$ and $n$ respectively, and if $h(x)=(f\circ g)(x)$, then the degree of $h$ is
   (1) $mn$ (2) $m+n$ (3) $m^n$ (4) $n^m$
3. A polynomial equation in $x$ of degree $n$ always has
   (1) $n$ distinct roots (2) $n$ real roots (3) $n$ complex roots (4) at most one root.
4. If $\alpha,\beta$, and $\gamma$ are the zeros of $x^3+px^2+qx+r$, then $\sum\frac{1}{\alpha}$ is
   (1) $-\frac{q}{r}$ (2) $-\frac{p}{r}$ (3) $\frac{q}{r}$ (4) $-\frac{q}{p}$
5. According to the rational root theorem, which number is not possible rational zero of $4x^7+2x^4-10x^3-5$?
   (1) $-1$ (2) $\frac{5}{4}$ (3) $\frac{4}{5}$ (4) 5

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